

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 3 Solutions

1. For $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ -1 & 3 & 6 \\ 2 & 1 & 0 \\ 3 & 1 & -1 \end{pmatrix}$, the row echelon form is $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

- (i) $\{(1, 2, 0), (0, 1, 1), (0, 0, 1)\}$ is a basis for the row space.
 $\{(1, 0, -1, 2, 3), (2, 1, 3, 1, 1), (0, 1, 6, 0, -1)\}$ is a basis for the column space.
 (Here (a, b, c, d, e) refer to the column form of the vector.)

Note: You can also perform Gauss Jordan elimination on $\mathbf{A}$ to get the reduced

row-echelon form: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

In this case, the basis for row space will be given by $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$, which is also a correct answer. This will not affect the answers for the other parts.

- (ii) This is the same as finding basis for the solution space of the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$. In this case, the system has only the trivial solution. So the basis for the nullspace is the empty set.
- (iii) $\text{rank}(\mathbf{A}) = 3$, $\text{nullity}(\mathbf{A}) = 0$.
 $\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = 3 = \text{number of columns of } \mathbf{A}$.

For $\mathbf{B} = \begin{pmatrix} 2 & 1 & 4 & 1 & 2 \\ 4 & 2 & 2 & 3 & 2 \\ 2 & 1 & -2 & 2 & 0 \\ 6 & 3 & 6 & 4 & 4 \end{pmatrix}$, the row echelon form is $\begin{pmatrix} 2 & 1 & 4 & 1 & 2 \\ 0 & 0 & -6 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

- (i) $\{(2, 1, 4, 1, 2), (0, 0, -6, 1, -2)\}$ is a basis for the row space.
 $\{(2, 4, 2, 6), (4, 2, -2, 6)\}$ is a basis for the column space.

(Using reduced row-echelon form: $\begin{pmatrix} 1 & \frac{1}{2} & 0 & \frac{5}{6} & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{1}{6} & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$, basis for the row

space is given by $\{(1, \frac{1}{2}, 0, \frac{5}{6}, \frac{1}{3}), (0, 0, 1, -\frac{1}{6}, \frac{1}{3})\}$.

- (ii) This is the same as finding basis for the solution space of the homogeneous system $\mathbf{B}\mathbf{x} = \mathbf{0}$.
 $\{(-\frac{1}{2}, 1, 0, 0, 0), (-\frac{5}{6}, 0, \frac{1}{6}, 1, 0), (-\frac{1}{3}, 0, -\frac{1}{3}, 0, 1)\}$ is the basis for the nullspace.

- (iii) $\text{rank}(\mathbf{B}) = 2$, $\text{nullity}(\mathbf{B}) = 3$.
 $\text{rank}(\mathbf{B}) + \text{nullity}(\mathbf{B}) = 5 = \text{number of columns of } \mathbf{B}$.

For $\mathbf{C} = \begin{pmatrix} 1 & 4 & 5 & 8 \\ -1 & 4 & 3 & 0 \\ 2 & 0 & 2 & 1 \end{pmatrix}$, the row echelon form is $\begin{pmatrix} 1 & 4 & 5 & 8 \\ 0 & 8 & 8 & 8 \\ 0 & 0 & 0 & -7 \end{pmatrix}$.

- (i) $\{(1, 4, 5, 8), (0, 8, 8, 8), (0, 0, 0, -7)\}$ is a basis for the row space.
 $\{(1, -1, 2), (4, 4, 0), (8, 0, 1)\}$ is a basis for the column space.

Using reduced row-echelon form: $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

the basis for row space is given by $\{(1, 0, 1, 0), (0, 1, 1, 0), (0, 0, 0, 1)\}$.

- (ii) This is the same as finding basis for the solution space of the homogeneous system $\mathbf{C}\mathbf{x} = \mathbf{0}$.

$\{(-1, -1, 1, 0)\}$ is a basis for the nullspace.

- (iii) $\text{rank}(\mathbf{C}) = 3$, $\text{nullity}(\mathbf{C}) = 1$.
 $\text{rank}(\mathbf{C}) + \text{nullity}(\mathbf{C}) = 4 = \text{number of columns of } \mathbf{C}$.

2. (i) Basis for row space $\{(1, 2, 3), (0, 0, 1)\}$. Take the nonzero rows of the r.e.f.
 (Answer not unique)

- (ii) Not possible, as we do not know the original matrix $\mathbf{A}$.

- (iii) Column space of $\mathbf{A}^T = \text{row space of } \mathbf{A}$. Same answer as part (i).

- (iv) Need to solve the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$, say with variables x, y, z .
 Using the r.e.f, we solve $z = t, y = t, x = -2t$.

Basis for the nullspace is $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$.

- (v) $\text{rank}(\mathbf{A}) = 2$ (two vectors in basis for row space); and $\text{nullity}(\mathbf{A}) = 1$ (one vector in the basis for nullspace).

3. If the three vectors are linearly independent, then they do not lie on the same plane.

Conversely, if the three vectors are linearly dependent, then they lie on the same plane (i.e. they are coplanar).

- (a) Use the three vectors to form a 3×3 determinant:

$$\begin{vmatrix} 2 & 6 & 2 \\ -2 & 1 & 0 \\ 0 & 4 & -4 \end{vmatrix} = -72$$

Since the determinant is non-zero, the three vectors are linearly independent, and hence they are not coplanar.

(b) Similarly, use the three vectors to form a 3×3 determinant:

$$\begin{vmatrix} -6 & 3 & 4 \\ 7 & 2 & -1 \\ 2 & 4 & 2 \end{vmatrix} = 0$$

Since the determinant is zero, the three vectors are linearly dependent, and hence they are coplanar.

4. (a) The general vector is $(a, b, a - b, a + b) = a(1, 0, 1, 1) + b(0, 1, -1, 1)$. Hence the subspace is $\text{span}\{(1, 0, 1, 1), (0, 1, -1, 1)\}$. Since the set of two vectors are linearly independent (note that $(1, 0, 1, 1), (0, 1, -1, 1)$ are not multiple of each other), it is a basis for the subspace. And the dimension is 2.
- (b) The general vector is $(a, a, a, a) = a(1, 1, 1, 1)$. Hence the subspace is $\text{span}\{(1, 1, 1, 1)\}$. $\{(1, 1, 1, 1)\}$ is a basis for the subspace. And the dimension is 1.

5. (i) Consider the system

$$\begin{array}{rcrcrcrcl} x & + & y & & & & = & 0 \\ & & & & y & + & z & = & 0 \end{array}.$$

Let $z = t$, so $y = -t$ and $x = t$. Then the general solution is $(x, y, z) = t(1, -1, 1)$.

The line of intersection of the two planes is given by $\text{span}\{(1, -1, 1)\}$.

(ii) The nullspace of the required $\mathbf{B}$ is the solution space of the homogeneous system $\mathbf{B}\mathbf{x} = \mathbf{0}$.

From part (i), we can take $\mathbf{B}$ to be $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ x & y & z \end{pmatrix}$, with the third row as any linear combination of the first two rows.

For example, you may simply take $\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

In general, $\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ a & a+b & b \end{pmatrix}$ for any a, b parameters.

6. (i) Consider the vector equations:

$$a(1, -1, 0, 0) + b(0, 1, 1, 0) + c(0, 0, 0, 1) = (0, 0, 0, 0)$$

which we can easily solved $a = b = c = 0$. Hence the vector equation has only the trivial solution.

We conclude that the vectors are linearly independent.

(ii) Let (a, b, c, d) be a vector that is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$. So

$$\begin{array}{rclcl} (a, b, c, d) \cdot (1, -1, 0, 0) & = & 0 & \Rightarrow & a - b = 0 \\ (a, b, c, d) \cdot (0, 1, 1, 0) & = & 0 & \Rightarrow & b + c = 0 \end{array}.$$

So such a vector is of the form $(a, a, -a, d)$ where a, d are parameters.

We may take the vector $(1, 1, -1, 0)$.

Note that this vector is not a multiple of $\mathbf{w} = (0, 0, 0, 1)$ and hence does not belong to $\text{span}\{\mathbf{w}\}$.

7. From the information given, we know the two solutions that the engineer found for homogeneous system $\mathbf{Ax} = \mathbf{0}$ are linearly independent and span the solution space of the system. Hence it is a basis for the solution space, which is also the basis for the nullspace of $\mathbf{A}$. Hence nullity $\mathbf{A} = 2$.

Since the matrix has 42 columns, by the Dimension Theorem, we deduce that $\text{rank } \mathbf{A} = 42 - \text{nullity } \mathbf{A} = 40$.

This implies all the 40 rows in the row echelon form of $\mathbf{A}$ are non-zero. So any non-homogeneous system $\mathbf{Ax} = \mathbf{b}$ will have a solution regardless of the vector $\mathbf{b}$.

8. (i) Under the given rotation,

$(1, 0, 0)$ is transformed into $(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$,

$(0, 1, 0)$ is transformed into $(-\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$,

while $(0, 0, 1)$ remains unchanged.

Under the given reflection,

$(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$ and $(-\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$ remain unchanged,

while $(0, 0, 1)$ is transformed into $(0, 0, -1)$.

So the basis that determine the new coordinate axes are $(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$, $(-\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$ and $(0, 0, -1)$.

- (ii) To find the coordinates c_1, c_2, c_3 relative to the basis S , we solve

$$c_1 \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix} \quad (*).$$

Solving the system, we get

$$c_1 = 2, c_2 = -2\sqrt{3}, c_3 = -3.$$

- (iii) From (*), we have the matrix equation form

$$\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -1 \end{pmatrix} (\mathbf{v})_S = \mathbf{v}$$

$$\text{So the required matrix } \mathbf{M} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

9. (i) Substituting the data pairs $(t, y) = (0, 0), (1, 1), (2, 3), (3, 6), (4, 10)$ into $y = a + bt + t^2$, we get:

$$\begin{array}{rcl} 0 & = & a + b \times 0 + 0^2 \\ 1 & = & a + b \times 1 + 1^2 \\ 3 & = & a + b \times 2 + 2^2 \\ 6 & = & a + b \times 3 + 3^2 \\ 10 & = & a + b \times 4 + 4^2 \end{array} \implies \begin{cases} a & = & 0 \\ a + b & = & 0 \\ a + 2b & = & -1 \\ a + 3b & = & -3 \\ a + 4b & = & -6 \end{cases}$$

$$\text{Set } \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ -1 \\ -3 \\ -6 \end{pmatrix}.$$

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}$$

$$\mathbf{A}^T \mathbf{b} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -1 \\ -3 \\ -6 \end{pmatrix} = \begin{pmatrix} -10 \\ -35 \end{pmatrix}$$

$$\text{Least squares solution: } \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{50} \begin{pmatrix} 30 & -10 \\ -10 & 5 \end{pmatrix} \begin{pmatrix} -10 \\ -35 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{3}{2} \end{pmatrix}.$$

So the equation of the quadratic curve: $y = 1 - \frac{3}{2}t + t^2$.

- (ii) At $t = 5$, the position $y = 1 - \frac{15}{2} + 25 = 18.5\text{m}$.
The velocity is given by $y' = -\frac{3}{2} + 2t$. At $t = 5$, $y' = 8.5\text{ms}^{-1}$.

10. (i) $\text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}.$

- (ii) $x - 2y + z = 0$. (Substitute $(2, 1, 0), (0, 1, 2)$ into linear equation $ax + by + cz = 0$ and solve for a, b, c .)

- (iii) Substituting $(1, 0, 0)$ into the equation in (ii) shows that $(1) - 2(0) + (0) \neq 0$. So it does not satisfy the equation, and hence $\mathbf{b}$ does not lie on the plane.

(iv) $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$ and $\mathbf{A}^T \mathbf{b} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$

The least squares solution:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{24} \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \frac{1}{12} \begin{pmatrix} 5 \\ -1 \end{pmatrix}.$$

(v) Projection $\mathbf{w} = \mathbf{A} \times (\text{least squares solution in (iv)}) = \begin{pmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 5/12 \\ -1/12 \end{pmatrix} =$

$$\frac{1}{6} \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}.$$

(vi) Note that $\begin{pmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 5/12 \\ -1/12 \end{pmatrix}$ in (v) can be written as

$$\frac{5}{12} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{12} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

So $\mathbf{w} = \frac{5}{12}\mathbf{v}_1 - \frac{1}{12}\mathbf{v}_2$.

Hence the coordinate vector is $\begin{pmatrix} 5/12, & -1/12 \end{pmatrix}$.

(vii) Shortest distance:

$$\|\mathbf{w} - \mathbf{b}\| = \left\| \left(\frac{5}{6}, \frac{1}{3}, -\frac{1}{6} \right) - (1, 0, 0) \right\| = \sqrt{\left(-\frac{1}{6} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(-\frac{1}{6} \right)^2} = \frac{\sqrt{6}}{6}.$$